STD 8 SOCIAL SCIENCE

SURE A PLUS NOTES

Movement of the Earth: Rotation and Revolution

1.What is Earth's rotation ?

It is the process by which the Earth spins on its own axis, an imaginary line running through its North and South Poles.

This rotation takes approximately 24 hours(23 hours 56 minutes 4 seconds).

The Earth rotates from west to east.So Sun rises in the east and sets in the west.

The main effects of the rotation of the Earth are day and night ,Tides and the Coriolis effect

2.What is the tilt of the Earth's axis ?

$ 23 1/2°

3.Define Circle of Illumination

@ The Earth receives light from the Sun.

@ During rotation, the part of the Earth facing the Sun has daytime and the other part experiences night.

@ The imaginary line that demarcates day and night on the Earth is called the Circle of Illumination.

@ It is not parallel to the Earth's axis.

4.Elucidate Coriolis Effect and Ferrel's law

• Due to rotation, freely moving bodies on the Earth's surface get deflected in their direction.

• The force responsible for this deflection is known as Coriolis Force and the deflection in direction is called Coriolis Effect.

• Admiral Ferrel discovered that due to the Coriolis Effect, ocean currents and winds change their direction in the northern hemisphere to the right and in the southern hemisphere to the left.

•This is known as Ferrel's law.

5.Briefly explain Revolution

➢ While rotating on its axis, the Earth revolves around the Sun in a fixed orbit.

➢ This is known as Revolution.

➢ The time taken to complete one revolution in the elliptical orbit is 365¼ days.

➢ 365 days is considered as one year for practical convenience.

6.What do you mean by leap year

☮ The fraction of 1⁄4 days is added once in 4 years to the February making it 29 days.

☮ Thus the year which has 366 days is called a leap year.

7.What is Perihelion and Aphelion ?

* The Earth's elliptical orbit causes varying distances from the Sun.

* It is closest at 147 million km during Perihelion around January 3, and farthest at 152 million km during Aphelion around July 4.

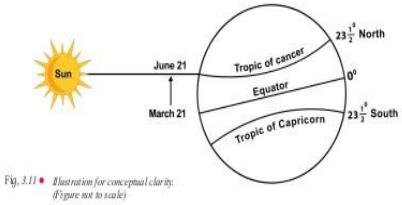
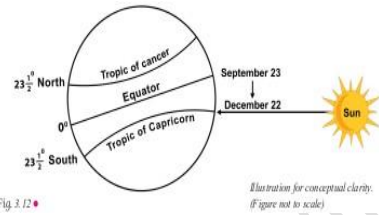
8.Elucidate Apparent Movement of the Sun (Third Term Model Qn Paper)

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@ Since the tilt of the earth's axis is maintained at an angle of $23\frac{1}{2}^0$ throughout the revolution.

@ The Sun's apparent position moves northward and southward between the Tropic of Cancer and the Tropic of Capricorn.

@ This apparent shift in the Sun's position is called the apparent movement of the Sun. For this reason, the duration of day and night also changes.

Day	Position of Sun	Peculiarity of the day
21st March (Spring Equinox)	The sun's rays fall vertically on the equator	Equinoxes-The duration of day and night will be equal on both hemispheres on these days
21st June(Summer solstice)  Fig. 3.11 Illustration for conceptual clarity. (Figure not to scale)	The Sun's apparent position shifts towards north from the Equator to the Tropic of Cancer from 21st March to 21st June (First Term Exam)	on 21st June, the northern hemisphere experiences the longest day and the shortest night
23rd September (Autumnal Equinox)	The sun's rays fall vertically on the equator	Equinoxes-The duration of day and night will be equal on both hemispheres on these days
 Fig. 3.12 Illustration for conceptual clarity. (Figure not to scale) 22 nd December(Winter Solstice)	Sun's apparent position shifts from the equator to the Tropic of Capricorn during the period from 23rd September to 22nd December. (First Term Exam) (Second Term Exam)	on 22 nd December, the southern hemisphere experiences the longest day and the shortest night

9. Compare Uttarayanam and Dakshinayanam

Uttarayanam	Dakshinayanam
# Following the winter solstice (December 22) the apparent movement of the Sun from Tropic of Capricorn ($23\frac{1}{2}^0$ South) to Tropic of Cancer ($23\frac{1}{2}^0$ North) is known as the apparent movement of the Sun towards North .	# The shift in the apparent position of the Sun after the Summer Solstice (June 21) from the Tropic of Cancer ($23\frac{1}{2}^0$ North) to Tropic of Capricorn ($23\frac{1}{2}^0$ South) is known as the apparent movement of the Sun towards South

10.What do you meant by Seasons ? List the reasons for Seasons

$ Due to the apparent shift in the position of the Sun, different places experience specific weather patterns.

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$ These patterns are known as seasons.
$ The revolution of the Earth and variations in the availability of solar energy is the reason for the occurrence of the seasons.
$ The occurrence of spring, summer, autumn and winter in a cyclical manner during a year is called seasonal change.

11.Tabulate different seasons and their characteristics (First Term Exam)

Seasons	Characteristics
Spring	• Plants bloom and produce fruits. • During this period duration of daytime gradually increases
Summer	• High atmospheric temperature .Generally longer days
Autumn	• Trees shed their leaves before the arrival of winter. • During this period duration of daytime gradually decreases
Winter	•Low atmospheric temperature. Snowfall generally longer nights

12. Tabulate the Cyclic Nature of Seasons

Apparent movement of the Sun	Duration	Northern Hemisphere	*Southern Hemisphere*
From the Equator to the Tropic of Cancer	March 21 to June 21	Spring	*Autumn*
From the Tropic of Cancer to the Equator	June 21 to September 23	Summer	Winter
From the Equator to the Tropic of Capricorn	September 23 to December 22	Autumn	Spring
From the Tropic of Capricorn to the Equator	December 22 to March 21	Winter	Summer

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13.What is Local Time ?
➢ When the sun is vertically overhead, it is noon.
➢ The time estimated at each place, based on the overhead position of the sun is termed as the local time.
➢ Thus the time calculated based on the length of the shadow and position of the Sun is termed as the Local time.
14. What is standard time? What is the need to calculate standard time in a country? (Second Term Exam)
⊙ The local time would be different at each longitude.
⊙ It would create a lot of confusion.

⊙ To overcome this crisis, based on an international understanding, countries have selected a longitude, which is a multiple of $7\frac{1}{2}^0$ longitude as Standard Meridian.
⊙ The local time at this Standard Meridian is considered as the standard time of the country.

15.Write a short note on Indian Standard Time

* There is around 30^0 longitudinal difference between India's easternmost state of Arunachal Pradesh and the westernmost state of Gujarat .
* So there is almost two hours of difference in the local time of these places.
* To avoid the practical difficulty caused by this, a standard time has been fixed for the country.
* The 82 1⁄2° longitude is considered as standard meridian of India.
* This is known as the Indian Standard Time (IST).

16.Elucidate Greenwich Mean Time (Third Term Model Qn Paper)

@ International time calculation, the longitude that passes through the Royal British Observatory in England is taken as zero degree meridian.
@ This is called Prime Meridian.
@ The local time at Prime Meridian is called the Greenwich Mean Time. Travellers worldwide calculate time based on Greenwich Time

17.Give a brief account on Time Zones

Based on the Greenwich Meridian, the world is divided into 24 zones, each with a time difference of one hour.
Each time zone has 15^0 longitudinal distance.
Countries like Russia, the USA, and Australia with large longitudinal extensions have many time zones and Standard Time.

18.What do you mean by International Date Line (First Term Exam)

$ 180^0 longitude is considered as the International Date Line.
$ The person travelling towards the east loses a day while crossing this longitude and a person travelling towards the west gains a day.
$ Travellers who move westwards crossing the line add a day and the travellers who move towards the east calculate time by deducting a day.
$ To avoid the situation of two different dates in countries through which the International Date Line passes, certain adjustments are made in this line.
$ This line is arranged to avoid populated land areas in the Pacific Ocean.

19.Local time at 15^0 West and 15^0 East when the Greenwich Mean time is 8 AM (First Term Exam)

15^0 West – 7 AM
15^0 East – 9 AM

20.Greenwich Mean time is 10 March 10 AM ;what will be time in 180^0 W and 180^0 E (Third Term Exam)

180^0 W 9 March 10 PM
180^0 E 10 March 10 PM

21. Greenwich Time 12.30 PM What will be time in (Third Term Exam)

150^0 E -10.30 AM

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150^0 W -2.30 AM

22.What will be the time at 1^0 East and 1^0 West when it is 10:00 A.M. at Greenwich Meridian (0^0 longitude)?

1^0 East – 10.04 AM

1^0 West – 9.56AM

23.When it is 9.00 P.M. at Greenwich Meridian (0^0 longitude) what will the time be at 30^0 East?

30^0 East- 11 P.M

All the Very Best Dear Students

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